

NAZARIO SALDAÑA

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EDUCATION

The University of Texas at Austin – Austin, TX

May 2028

Bachelor of Science in Electrical & Computer Engineering

Relevant Coursework: Computer Architecture, Embedded Systems, Digital Logic Design, Circuit Theory, Algorithms

PROJECTS

ASCEND ATV – *Embedded Firmware*

September 2025 - Current

- Developed firmware for synchronized dual-actuator control preventing hardware damage via system calibration
- Created detailed wiring schematics and calibration documentation to streamline team-wide system debugging
- Implemented a closed-loop switch to establish hardware safety methods and prevent electrical failure by 100%

Wearable Fitness Tracker – *C, Embedded Systems*

September 2026 – Current

- Integrated heart rate, SpO2, and 6-axis motion sensor to capture continuous physiological and motion data
- Developing custom C firmware for real-time signal processing and low-power BLE streaming on compact Li-Po cell
- Designed custom PCB for assembly, including power management and sensor routing
- Building a native iOS app in Swift that receives and displays heart rate, activity, and various health metrics

LC-3b Assembly Machine Architecture – *C, Computer Architecture*

August 2026 - Current

- Developed a 2-pass assembler to translate LC-3b assembly language source code into machine language
- Designed an instruction-level simulator to model the LC-3b microarchitecture and execute the assembled machine code
- Engineering a microarchitecture-level simulator by developing custom microcode to execute each LC-3b instruction

Space Invaders Game Development – *C, Embedded Systems*

November 2025

- Built a Space Invaders game using interrupt system logic for real-time input, custom graphics, and audio integration
- Designed modular firmware including collision detection, event timing, and real-time player score system
- Implemented interactive spaceship movement on a 128x160 LCD screen using a slide potentiometer as an analog input

LEADERSHIP & COMMUNITY INVOLVEMENT

Society of Hispanic Professional Engineers – Austin, TX

May 2026 – Current

Project Development Chair

- Created 9+ project events with the purpose of improving member technical expertise across all engineering disciplines
- Hosted 5+ panel events with industry professionals to promote networking and recruitment opportunities for members
- Collaborated with corporate sponsors to organize 3+ events, including a yearly Makeathon initiative with Microsoft

Recruitment & Retention Co-Chair

May 2025 – May 2026

- Organized 15+ events, including a themed cookout with food and games to increase interaction by 28%
- Boosted membership by 25% by fostering an inclusive environment through outreach and personal engagement
- Led Manitos & Manitas mentorship for 100+ members, increasing new member activity by 50% YoY
- Managed game logistics for 100+ members, including partner pairings, target data, and real-time communications

Ramshorn Mentor – Austin, TX

August 2025 – Current

Academic Coach

- Mentored 8 first-year Ramshorn Scholars, supporting academic, social, and personal transition into college
- Tutored in challenging subjects to strengthen academic performance and develop effective study strategies
- Supported professional growth via career resources, reviewing resumes, and facilitating networking opportunities

DELL Technologies STEM Aspire Program

August 2026 - Current

Participant

- Selected as 1 of 38 students to participate in STEM Aspire, to accelerate Career readiness and professional growth
- Engaged in 1:1 mentorship and career coaching with corporate leaders to navigate professional pathways
- Strengthened technical and professional competencies via industry workshops, networking, and site visits

TECHNICAL SKILLS

- **Languages:** C/C++, ARM Assembly, Verilog, Python, Java
- **Software:** VSCode, Xilinx Vivado, KiCad, LTspice, AutoCAD, Autodesk Inventor, Revit, MATLAB
- **Hardware:** Oscilloscope, Microcontroller, Raspberry Pi, Arduino, Solder Iron, FPGA

HONORS

Ramshorn Scholar | HSF Scholar | University Leadership Network Scholar | Cockrell Engineering Scholarship Recipient